

$GF(2^8)$

The encryption algorithms used in the Advanced Encryption Standard (AES) operate on 8-bits of data at a time and require a finite field of order 256. This is accomplished with 256 polynomial elements that are multiplied and added modulo $m(x) = x^8 + x^4 + x^3 + x + 1$.

These 256 polynomial elements are represented in hex as: 00, 01, 02, 03, ... , F9, FA, FB, FC, FD, FE, FF.

Addition and multiplication are done directly on the 8-bit binary strings using:

Addition: Bitwise $\oplus$

Multiplication by 00000010 :

- 1-bit left shift $(b_6b_5b_4b_3b_2b_1b_00)$ if $b_7 = 0$
- 1-bit left shift $(b_6b_5b_4b_3b_2b_1b_00) \oplus (00011011)$ if $b_7 = 1$

Note: The XOR bit pattern (00011011) corresponds to:

$$\begin{aligned} m(x) - x^8 &= x^8 + x^4 + x^3 + x + 1 - x^8 \\ &= x^4 + x^3 + x + 1 \end{aligned}$$

Examples:

$$\begin{aligned} \{5A\} + \{02\} &= (01011010) + (00000010) \\ &= (01011000) \\ &= \{58\} \end{aligned}$$

$$\begin{aligned} \{5A\} \times \{02\} &= (01011010) \times (00000010) \\ &= (10110100) \\ &= \{B4\} \end{aligned}$$

$$\begin{aligned} \{80\} \times \{02\} &= (10000000) \times (00000010) \\ &= (00011011) \\ &= \{1B\} \end{aligned}$$

Exercises:

Evaluate the following additions and multiplications using 8-bit binary in $\text{GF}(2^8)$:

$$\{3C\} + \{9A\}$$

$$\{F0\} + \{11\}$$

$$\{70\} \times \{02\}$$

$$\{70\} \times \{04\}$$

$$\{D4\} \times \{02\}$$

$$\{6A\} \times \{03\}$$

Answers:

Evaluate the following additions and multiplications using 8-bit binary in $GF(2^8)$:

$$\{3C\} + \{9A\} \text{ A6}$$

$$\{F0\} + \{11\} \text{ E1}$$

$$\{70\} \times \{02\} \text{ E0}$$

$$\{70\} \times \{04\} \text{ DB}$$

$$\{D4\} \times \{02\} \text{ B3}$$

$$\{6A\} \times \{03\} \text{ BE}$$